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Review

Application of machine learning in the design of biomedical nanomaterials

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ABSTRACT

As artificial intelligence technology advances rapidly, machine learning shows great potential in materials science. It can efficiently process and analyze large-scale data, uncover complex patterns, and offer new approaches for material prediction and design. This review focuses on the application progress of machine-learning models in biomedical nanomaterial design. Firstly, it presents the basic principles, development history, and advantages of machine learning in this field. Then, it provides a deep analysis of the technical breakthroughs in machine-learning models for predicting physicochemical properties, optimizing synthesis routes, and analyzing biological interactions of nanomaterials. Finally, it addresses the challenges of insufficient data quality and limited model interpretability in machine learning, and proposes solutions like developing more explainable machine-learning methods, standardizing data-reporting formats, combining machine learning with traditional molecular simulations, and enhancing result verification. Overall, this review aims to promote a shift from experience-driven to data-driven approaches in the field of biomedical nanomaterial design and to facilitate innovation and clinical translation in this area.

1. Introduction

Biomedical nanomaterials refer to materials with nanoscale size, which can realize the functions of diagnosis, treatment, or tissue repair in the biomedical field through specific design and functional modification [1]. A notable example is liposomes, which marked the beginning of modern biomedical nanomaterials and were first developed by Bangham in 1965 [2]. In 1995, the FDA approved the first nano-drug, Doxorubicin liposome (Doxil), demonstrating that PEGylated liposomes significantly increased the tumor accumulation and anti-tumor efficacy of doxorubicin in animal models while reducing cardiotoxicity [3]. The approval was based on data from a multicenter, open-access, key phase II/III clinical trial of Doxil (then known as DOX-SL) in the treatment of patients with AIDS-related Kaposi's sarcoma [4]. Studies confirmed that Doxil's high therapeutic efficacy (objective response rate $>$ 80%) and its substantially lower side effects, such as cardiotoxicity and myelosuppression, compared to conventional doxorubicin. In 2018, liposome-encapsulated siRNA technology targeting transthyretin emerged [5], becoming the world's first approved siRNA drug. Liposomes solved the core problems of the instability of siRNA in vivo and the difficulty of targeted delivery to hepatocytes, achieving the clinical application of RNA interference therapy. Through their unique physicochemical properties and interactions with biological systems, these materials have significantly improved the efficiency and safety of disease diagnosis and treatment, demonstrating their great potential in modern medicine. Beyond liposomes, diverse nanomaterial platforms have achieved remarkable clinical success. Albumin-bound paclitaxel nanoparticles (Abraxane) were approved by the FDA in 2005 for metastatic breast cancer, demonstrating superior efficacy of protein-based nanocarriers [6]. Polymeric nanoparticles such as polylactic-*co*-glycolic acid (PLGA) have enabled controlled drug release and are currently evaluated in multiple clinical trials for cancer therapy [7–9]. Iron oxide nanoparticles have been approved as MRI contrast agents (e.g., Feridex) and for magnetic hyperthermia treatments [10]. Dendrimers, with their precise structure, facilitate targeted drug delivery and gene transfection [11]. These advances collectively underscore the breadth of

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nanomaterial innovations transforming modern medicine.

Despite significant advances, traditional biomedical nanomaterials face growing limitations. Traditional studies rely on trial-and-error methods and high-throughput experimental screening, which are not only time-consuming and labor-intensive, but also difficult to comprehensively analyze the complex correlation among multiple parameters [12,13]. For example, when optimizing the synthesis process of lipid nanomaterials, numerous variables such as lipid composition, particle size, and surface charge need to be regulated simultaneously [14]. Traditional methods often fail to identify the optimal conditions efficiently due to the combinatorial explosion of parameters. Moreover, a deeper challenge arises from the limited understanding of the interaction between biomedical nanomaterials and biological systems. Despite the boom in nanocarrier research since 1965, only six nanocarrier types have been approved by the FDA for clinical use, and the key bottleneck is low drug delivery efficiency [15]. This inefficiency is rooted in the fact that the dynamic linkages from the tissue and cellular to the molecular level have not been fully resolved. Drug carriers must overcome multiple transport obstacles in vivo, including immune clearance, tissue penetration barriers, insufficient cellular uptake efficiency, subcellular localization bias, non-specific biodistribution, and unclear metabolic pathways [16]. In addition, fragmented and non-standardized recording of experimental data further limits the reproducibility and universality of the results, prolongs the research and development cycle, increases costs, and ultimately restricts clinical translation [17].

The application of machine learning in materials science provides solutions to these challenges. By efficiently processing and analyzing large datasets, machine learning can reveal patterns that are difficult to identify through traditional experimental operations [18]. Unlike traditional methods, machine learning can automatically learn rules, quickly mine nonlinear relationships and make predictions, reduce the experimental workload and errors, and shorten the research and development cycle, thereby effectively breaking through the bottlenecks of traditional research and facilitating the rapid development of biomedical nanomaterials [19–21]. Machine learning is mainly classified into supervised learning (e.g., random forest, neural network, etc.) and unsupervised learning (e.g., clustering, principal component analysis, etc.). Since the perceptron model was proposed in the 1950s, machine learning has evolved from simple linear models to deep neural networks [22]. For example, random forest model can efficiently predict the adsorption behavior of nanomaterials by integrating multiple decision trees [23]. Deep learning models can extract abstract features from high-dimensional data to guide the reverse design of core-shell nanomaterials with specific optical properties [24].

Recently, remarkable progress has been made in the application of machine learning to the design of biomedical nanomaterials. The application of machine learning in predicting the biological activity and toxicity of nanomaterials [25], structure prediction [26–29], drug loading capacity prediction [25], and compound property prediction [30] has been systematically explored. However, there remains a lack of comprehensive reviews that systematically sort out the principles, development history, strengths, and weakness of various machine learning models, as well as their application in biomedical nanomaterials design. Therefore, this review aims to summarize the basic principles of machine learning models and their recent advancement in the design of biomedical nanomaterials, focusing on their technical breakthroughs in the prediction of physical and chemical properties, synthesis pathway optimization, and biological interaction analysis (Fig. 1). Furthermore, to address the challenges such as insufficient data quality and limited model interpretation, this review proposes a solution strategy combining physical models and experimental verification. It also outlines the future development direction, including standardized databases and cross-scale data fusion, to provide theoretical reference for promoting the transformation from experience-driven to data-driven in this field.

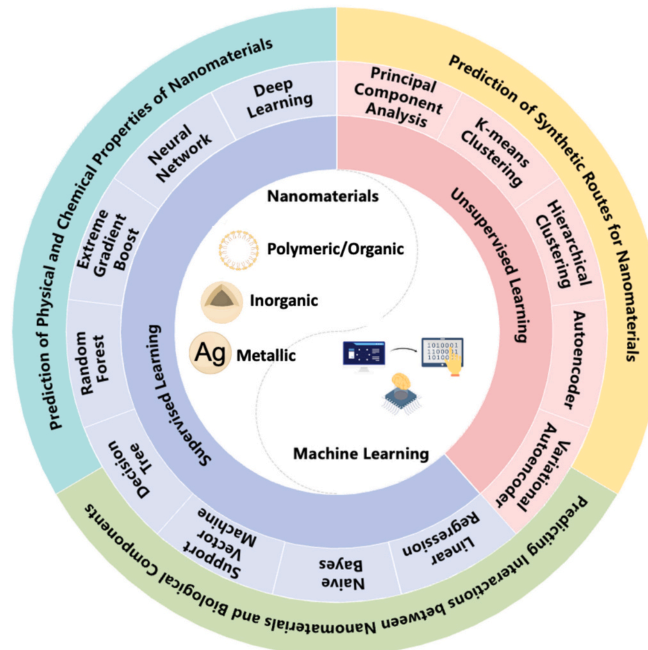

Fig. 1. Schematic overview of the review.

2. Models of machine learning

2.1. Unsupervised learning

Unsupervised learning is an algorithm that does not rely on the guidance of labeled data but autonomously discovers hidden patterns and relationships through the intrinsic structure and distribution of the data. It does not require explicit target variables or labels, but instead discovers patterns and structure from the data itself. The main strength of unsupervised learning lies in its ability to deal with large amounts of unlabeled data and discover underlying structures and relationships. However, the results are usually less interpretable and more sensitive to the quality and distribution of the data. While unsupervised learning can adapt to complex data distributions and discover hidden patterns and structures, it lacks a clear objective function, and its decision-making process is difficult to understand intuitively. In addition, unsupervised learning is more sensitive to data noise and outliers. This approach includes a variety of machine learning models, such as principal component analysis, K-means clustering, hierarchical clustering, autoencoder, and variational autoencoder, etc., each offering distinct features and strengths in data dimensionality reduction, cluster analysis, and feature learning (Table 1).

2.1.1. Principal component analysis

Principal component analysis (PCA) is an unsupervised machine learning method for dimensionality reduction of high-dimensional data through orthogonal transformations, with the core objective of extracting the main features in the data and eliminating redundant information, while maximizing the data variance to retain key information [31]. PCA simplifies the data structure by constructing new orthogonal variables.1 Principal components refer to linear combinations of the original variables and are ranked in descending order of variance to enable downscaling and feature extraction from the data. PCA is known for its mathematical simplicity and is suitable for processing multivariate and high-dimensional data in biomedical materials research, such as the correlation analysis of material performance parameters and feature extraction of high-throughput experimental data [32]. The advantage of PCA is that by projecting high-dimensional data into a low-dimensional space, the number of features is reduced, enabling significant reductions

Table 1
Introduction to models of unsupervised learning and their advantages and disadvantages.

Name	Brief intro	Advantages	Limitations
Principal component analysis (PCA)	Dimensionality reduction via orthogonal transforms	Handles high-dim data Noise reduction	High pre-processing Parameter sensitivity
K-means clustering	Divides data into k clusters Iterative optimization	Low cost No scaling needed	Initial center sensitivity Preset k
Hierarchical clustering	Tree-based clustering via distances	No preset k Flexible metrics	Computationally heavy Noise sensitive
Autoencoder	Encoder-decoder mapping to low-dim space	Feature extraction Denoising	Large data needed Overfitting risk
Variational autoencoder (VAE)	Generative model with variational inference	Continuous latent space Generates samples	Complex architecture Long training

in computation and lower storage requirements. In addition, PCA filters out random noise and redundant features from the data, thus enhancing model stability. Principal components can often be interpreted as linear combinations of primitive features, making PCA valuable in domain analysis: in genetic data analysis, principal components can reveal major patterns of gene expression; in image processing, principal components can capture key features of an image [32].

2.1.2. K-means clustering

While PCA provides a foundational approach to simplifying high-dimensional data by extracting key linear features, researchers often need to explore the intrinsic grouping structures within the data, which leads to the application of clustering methods. K-means clustering is a classical clustering algorithm whose core objective is to divide the dataset into k non-overlapping clusters through iterative optimization, such that the similarity of data points within each cluster is maximized and the similarity between different clusters is minimized [33,34] (Fig. 2). The K-means clustering algorithm continuously optimizes the cluster center locations by iteratively minimizing the total distance (i.e. sum of squared errors) between the samples in the cluster and their centroids to achieve data segmentation. The advantage of this algorithm is that the structure is simple. The final result is to divide the data points into k clusters [35]. The small number of iterations makes the computational complexity low, which is suitable for the initial feature classification and pattern recognition of large-scale data [33]. PCA, K-means clustering algorithm does not require data pre-processing for dimensionality reduction. It can process the original high-dimensional data directly, avoiding losing information during dimensionality reduction. In addition, K-means clustering is computationally efficient and suitable for large-scale datasets, retaining the interpretability of the original variables. In contrast, PCA is mainly used for dimensionality reduction. It assumes that the data are linearly divisible. PCA is ineffective for data with nonlinear relationships. Therefore, in the multivariate analysis of biomedical materials, K-means clustering algorithm has higher utility and applicability in optimizing material performance parameters and mining component correlations.

2.1.3. Hierarchical clustering

Hierarchical clustering is a method of cluster analysis that was developed from traditional algorithms such as K-means clustering [36]. Although K-means clustering is efficient for partitioning data into predefined groups, its requirement for a pre-specified number of clusters (k) can be a limitation when the underlying data structure is unknown. For more exploratory and hierarchical insights, hierarchical clustering offers a flexible alternative. This method gradually merges or splits data points by calculating the distance or similarity between them, eventually forming a dendrogram, which is used to reveal the similarity relationships of data at multiple scales. Similar to K-means clustering, the goal of hierarchical clustering is to divide the data into clusters such that the similarity of data within each cluster is maximized and the similarity of data between different clusters is minimized. Unlike K-means clustering, hierarchical clustering optimizes the calculation of the distance or similarity between data points by supporting multiple spacing measures, which makes it suitable for different types of analytical data. In addition, hierarchical clustering does not require a predetermined number of clusters, but rather presents the hierarchical structure of the data visually through a tree diagram, providing analysts with a decision basis for the number of clusters. This feature offers hierarchical clustering unique advantages in exploratory data analysis. For example, in outlier detection or feature correlation analysis, hierarchical clustering can clearly illustrate the hierarchical relationship of the data through a tree diagram, which helps researchers discover the hidden patterns and connections. In contrast, K-means clustering may have limitations in scenarios with unknown data structures due to the need to pre-specify the number of clusters. Therefore, hierarchical clustering can provide a more flexible and intuitive analysis perspective when dealing with complex datasets.

2.1.4. Autoencoder

Traditional methods like PCA, K-means, and hierarchical clustering often rely on linear assumptions or distance metrics. To capture more complex, nonlinear relationships within high-dimensional biomedical data, approaches such as autoencoders have gained prominence. An autoencoder is a model that aims to reconstruct data by learning a compressed representation of the data, and its basic structure consists of an encoder and a decoder. The encoder is responsible for mapping the input data to a low-dimensional latent space representation, while the decoder reconstructs the input data from that latent representation [37]. During training, the goal of the autoencoder is to minimize the error between the input and the reconstructed output, such as the mean

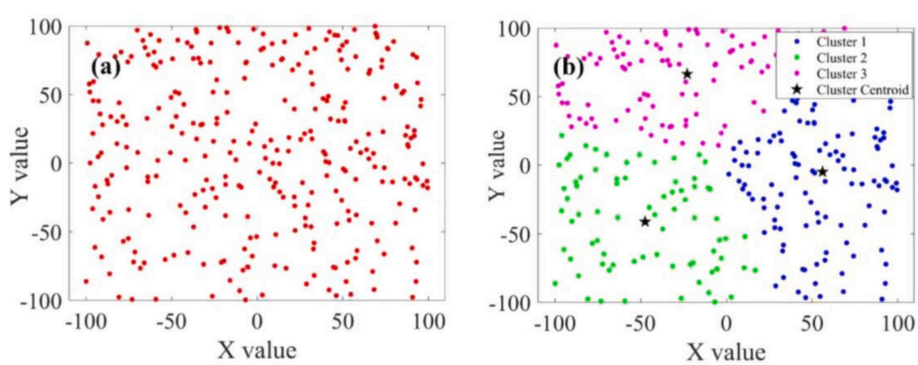

Fig. 2. K-means clustering. (a) a set of datasets and (b) closest cluster centroid with K = 3 [34].

square error or cross-entropy. This allows the model to learn the intrinsic feature representation of the data. A notable feature is the autoencoder's ability to automatically extract efficient features from the data, eliminating the need for manual feature design, thereby greatly reducing feature engineering effort. Autoencoder possesses powerful nonlinear modeling capabilities and is capable of capturing complex relationships in data. In addition, the autoencoder is versatile and suitable for a wide range of tasks, such as dimensionality reduction, data compression, denoising, and anomaly detection. It excels in data compression and visualization, facilitating storage and transmission, while also supporting visual analysis of data. Autoencoders are also flexible and scalable, and can be adapted and expanded to meet specific needs.

2.1.5. Variational autoencoder

Standard autoencoder is powerful for learning compressed representations and denoising, yet its latent space lacks a structured probabilistic framework, limiting its generative capability. To address this, the variational autoencoder (VAE) introduces a probabilistic approach to enable both robust representation learning and data generation. VAE is a autoencoder-based generative model that learns the latent representation of data by introducing variational inference [38]. VAE uses an encoder to map the input data to the parameters of the latent space (mean and variance) and then generates new data samples through a decoder. Properties of VAE include enhanced generative capabilities, potential spatial continuity, and interpretability. Through the combination of probabilistic coding spaces and regularization terms, VAE can randomly sample from the latent space to generate new samples with similar distributions to the original data, thus enhancing the diversity and realism of the generated data. In addition, the latent space in a VAE is continuous and interpretable, with distances between data points reflecting their similarity, which supports data visualization and interpolation [38]. VAE is an extension of the traditional autoencoder. While traditional autoencoder compresses the input data into a low-dimensional representation through an encoder and reconstruct it using a decoder, primarily for feature extraction and denoising tasks, to the VAE can also learn the latent distribution of the data and also generates new samples similar to the training data, demonstrating powerful generative capabilities. In anomaly detection, the traditional autoencoder relies only on the reconstruction error for judgment, which has a relatively limited effect. Instead, VAE can identify anomalous samples more accurately using probabilistic modeling of the latent space.

2.2. Supervised learning

Supervised learning is a method of training a model using known input and output data to make predictions or classify unknown data through pairwise correspondences that enable the model to grasp the mapping between input data and output labels [39]. The training process can be analogized to an experienced teacher guiding a student, allowing the model to gradually grasp the laws and patterns in the data by constantly providing examples of the correct answers. Supervised learning requires large amounts of labeled data to train the model, which acts as a priori knowledge to establish accurate correspondences for the model. Model performance can be quantified and optimized by metrics such as accuracy, recall, and F1 score. In addition, supervised learning algorithms are usually highly interpretable and can clearly demonstrate the prediction process and rationale. Both supervised and unsupervised learning models assume that the data obeys certain distribution and optimize the objective function by learning the distributional properties of the data. However, unsupervised learning does not rely on labels and aims to mine the intrinsic structure or patterns of the data, such as clustering or dimensionality reduction, whereas supervised learning relies on labels for prediction. Supervised learning has a wide range of applications in fields such as image recognition, disease diagnosis, and drug discovery, and its performance is relatively stable. Supervised learning encompasses a variety of models, such as linear regression, naive Bayes, support vector machines, decision trees, random forest, extreme gradient boosting, neural network, and deep learning (Table 2).

2.2.1. Linear regression

Linear regression is the simplest model in supervised learning and is mainly used for regression tasks, uncovering the correspondence between the independent and dependent variables [40]. The model predicts the dependent variable based on the new independent variable and its function that represents this mapping relationship. The basic form assumes a linear relationship between the dependent variable y and the independent variable x, i.e. $y = w^T x + b$, where w is the weight vector and b is the bias term. By fitting this model to the training data, the optimal values for w and b can be obtained so that the error between the predicted and true values is minimized. Linear regression models show good adaptability to small datasets and can be used to make effective predictions by fitting linear relationships even with limited data. The model also provides feature importance coefficients. By examining the magnitude of the weights corresponding to each feature in w, it is possible to understand how much each feature contributes to the prediction goal, which helps feature selection and understanding the relationship within the data. The ability to make predictions and provide clear feature importance coefficients makes linear progression one of the most fundamental algorithms in machine learning [40]. The fundamental limitation of linear regression lies in its inherent linear assumption, which constrains its applicability to real-world data. Primarily, the model cannot capture nonlinear relationships. This structural rigidity is compounded by its high sensitivity to outliers, which can disproportionately distort the model's fit. Furthermore, linear regression relies on a set of often unrealistic statistical assumptions—such as feature independence and homoscedasticity—and lacks any built-in mechanism for feature selection or handling multicollinearity. Consequently, when faced with high-dimensional data, it is prone to overfittings. In essence, while linear regression provides a transparent and interpretable baseline, its effectiveness is bounded by the linearity of the underlying phenomenon and the strictness of its foundational assumptions.

2.2.2. Naive Bayes

While linear regression establishes a foundational approach for understanding variable relationships in regression tasks, many cases in biomedical materials, such as material classification or toxicity grading,

Table 2
Introduction to models of supervised learning and their advantages and disadvantages.

Name	Brief intro	Advantages	Limitations
Linear regression	Models linear relationships	Simple Interpretable	Poor with non-linear data
Naive Bayes	Probabilistic classification (Bayes)	Fast Tolerates missing data	Independence assumptions
Support Vector Machine (SVM)	Kernel-based hyperplane optimization	Small sample performance	Kernel-dependent Large data slow
Decision Tree	Tree-based splits (yes/no)	Interpretable Visualizable	Over fits Noise sensitive
Random Forest	Ensemble of decision trees	Robust High accuracy	Black-box Sensitive to noise
Extreme Gradient Boost (XGBoost)	Gradient-boosted trees Loss optimization	Fast on big data Accurate	Memory-heavy Complex
Neural Network	Layered neuron simulation	Handles non-linear Fault-tolerant	Resource-heavy Slow training
Deep Learning	Multi-layer networks Non-linear transforms	Auto-features High-level semantics	Data-hungry Compute-heavy

require dedicated classification algorithms. Probabilistic models like Naive Bayes provide a direct solution for such scenarios. Naive bayes is a probabilistic classification algorithm based on Bayes' theorem, which introduced the idea of probabilistic classification, paving the way for advanced methods such as probabilistic graphical models [41]. The core idea is to classify a sample by calculating the posterior probability of each category, and selecting the category with the largest posterior probability as the classification result for that sample [42]. The posterior probability is the probability recalculated after accounting for a portion of the data that is known. The algorithm assumes that the features are independent, do not interfere with each other, and are not correlated[51]. Naive Bayes introduces the idea of probabilistic classification and offers support for more advanced methods such as probabilistic graphical models. It only adjusted three to four parameters, such as the smoothing parameter and prior probability, and the default values (e.g., smoothing parameter = 1.0) also perform well in most cases, which simplifies the model construction process. Another advantage of Naïve Bayes is its tolerance for missing values in the data set. Naïve Bayes implementations can naturally handle missing values at prediction time by excluding missing features from the probability calculation. This arises from the algorithm's factorized probability structure rather than any special robustness. However, this approach requires the missing-at-random assumption and may perform poorly when missingness is informative or when critical features are frequently absent. Datasets in biomedical and material science domains are usually not valid for the independence test due to more or less correlation between features, and it is possible to combine Naive Bayes with other classifiers to make use of their respective strengths and compensate for the shortcomings of this model, e.g., by projecting the features into a new space through methods such as PCA, so that the correlation between the features will be reduced [43].

2.2.3. Support vector machine

The assumption of feature independence in Naive Bayes can be a significant limitation for complex biomedical datasets where features are often correlated. To achieve more robust classification with complex decision boundaries, especially in high-dimensional spaces, support vector machines (SVM) offer a powerful alternative. SVM is a supervised learning model based on statistical learning theory. The core idea is to project data into a high-dimensional space using kernel functions to find the optimal linear hyperplane to distinguish between different classes of sample data (Fig. 3) [44], provide ideas for solving nonlinear problems by finding optimal hyperplanes and using kernel functions to deal with nonlinear problems [45]. Linear regression fits the linear relationship between input features and output by minimizing the sum of squared errors, which provides a theoretical basis for SVM when dealing with linearly differentiable cases. SVM addresses nonlinear problems by identifying optimal hyperplanes and utilizing kernel functions, which offer new approaches for solving complex nonlinear problems. Its advantage lies in its ability to transform nonlinear real-world problems into linearly differentiable problems, thus effectively handling high-dimensional data. In addition, the model performs well in handling small sample data and can effectively avoid overfitting.

SVM is a powerful supervised learning model that excels at handling high-dimensional data, preventing overfitting, and managing small datasets, but are less efficient when dealing with large-scale data. SVM is particularly advantageous in the biomedical field, especially in medical image segmentation, cancer diagnosis, and material property prediction. For example, in bioinformatics, the relationship between the structure and function of proteins and amino acid sequences is complex and nonlinear [46,47]. Researchers can collect amino acid sequences of proteins with known structures as training data. Each sequence can be represented as a feature vector, and the features may include the physicochemical properties of the amino acids, such as hydrophobicity, charge, and volume [48]. SVM then uses these feature vectors and the corresponding protein structural labels (e.g., α-helix, β-fold, or random curl) to train a classification model.

2.2.4. Decision tree

Despite its strength in finding optimal separating hyperplanes, SVM's decision process can be less interpretable as a “black box.” For scenarios where model transparency and the logic behind each prediction are crucial, tree-based models like decision trees provide a highly interpretable framework. A decision tree, as the name suggests, is a tree-structured model [49]. It asks a series of “yes or no” questions to determine which category the data belongs to, and gradually subdivides it into different subsets, like a tree with many branches. Eventually, the data set is divided into several subsets, each corresponding to a category. The similarity between decision trees and SVM is that they both divide the dataset into different subsets to achieve classification or regression tasks by means of feature segmentation or hyperplane division. The difference is that decision trees construct a global classification by creating several local divisions, with more classifications and more detailed divisions. In contrast, SVM directly slices up a large amount of data at the global level, making these subsets more differentiated as a whole and increasing the likelihood of individual misclassification [50].

With the increasing application of machine learning techniques in materials science, decision trees have been gradually refined and optimized as a base model for the design and performance prediction of biomedical materials. Decision trees are highly interpretable: their tree structure is similar to the human logical reasoning process, allowing user to follow the decision paths from top to bottom and left to right like reading a flowchart, and visualizing how each decision is divided based on features [51]. The theoretical training time complexity of decision tree models is O(m·n·log n) (for continuous features) or O(m·n) (for discrete features), where m represents the number of features and n

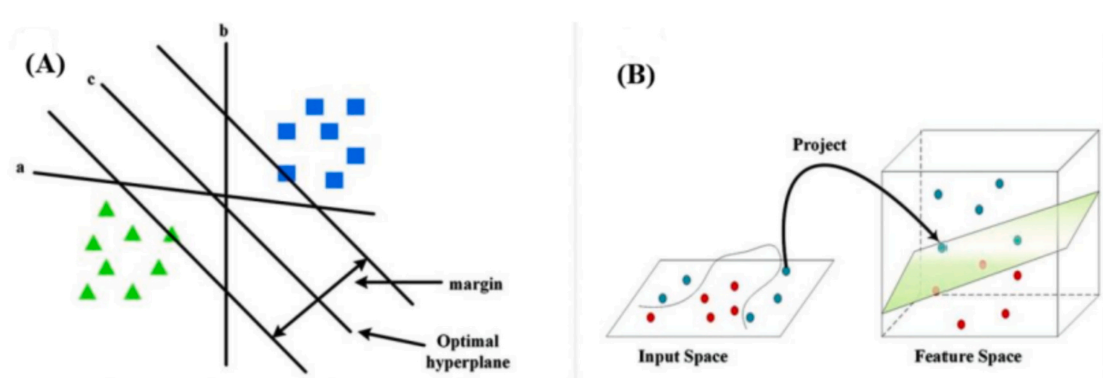

Fig. 3. The pattern diagrams of SVM. (A) Support vector machine (SVM) schematic. (B) SVM project data from low-dimensional to high-dimensional space and the determination of the hyperplane for classification [44].

represents the number of samples. Empirical research shows that although ensemble methods such as random forests have the highest accuracy, their computational complexity is also significantly higher than that of other algorithms. In a single-GPU environment, the training time for a medium-sized dataset with 68 K samples is approximately 30 to 100 min [52]. At the same time, they are constrained by hardware factors such as memory bandwidth and communication latency. Even with distributed training, it is still not feasible to complete the training within three months when the computational load exceeds 2×10^{28} flops. Therefore, the training time not only depends on the scale of the data, but is also closely related to the feature dimension, tree depth, hardware configuration and parallel strategy [53].

2.2.5. Random forest

Random forest is an integrated learning method that makes predictions by building multiple decision trees and integrating their results. In classification tasks, the final prediction is determined by 'collective decision making' (i.e., counting the most votes), while in regression tasks, the final result is the average of predictions of all the decision trees. A decision tree is a tree-structured model in which the decision-making process is represented by a series of decision nodes and branches, with each node representing a feature, and the branches representing the feature values, leading to the final result or regression value. During training, the model randomly selects different subsets from the original dataset by bootstrapping to ensure that each tree learns from different samples, thus avoiding overfitting (Fig. 4) [54]. In addition, when constructing a decision tree, random forest not only selects samples randomly, but also limits the number of features considered at each node. This strategy avoids over-reliance of the model on certain features and improves diversity and robustness. The advantages of random forests include their adaptability to data, ability to handle high-dimensional data, and effectiveness in capturing interactions between

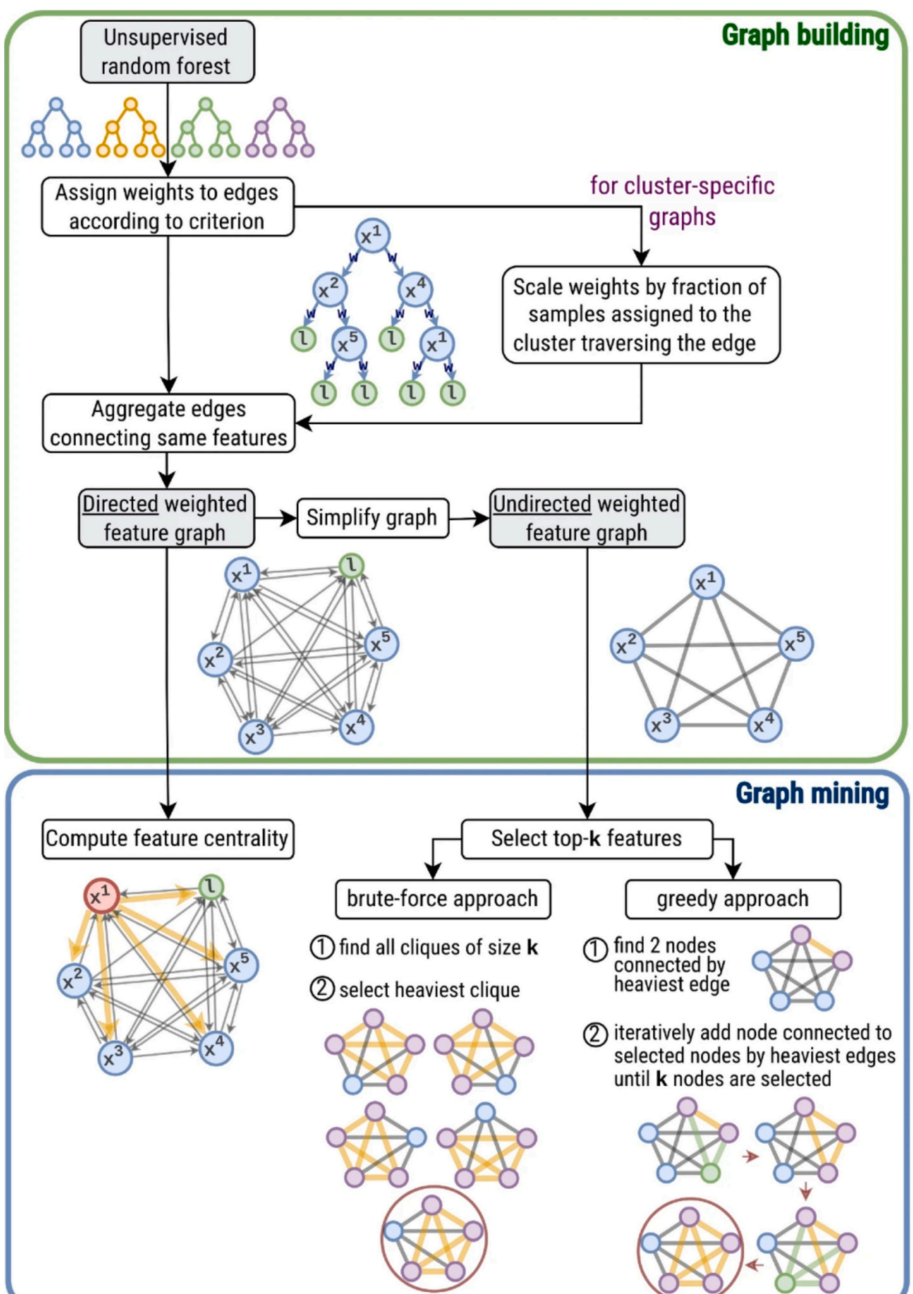

Fig. 4. The pattern diagram of the random forest [54].

features. In addition, it can assess the importance of features during the training process and provide a scientific basis for feature selection. This property makes it particularly effective for complex datasets, especially when the number of features is high or there are complex relationships between features.

2.2.6. Extreme gradient boosting

A further advancement in tree ensembles involves building trees sequentially, where each new tree corrects the errors of the previous ones, leading to powerful models like extreme gradient boosting (XGBoost). XGBoost is a gradient-based enhancement method that optimized decision tree and random forest by minimizing prediction error to improve prediction performance [55]. The decision tree contains leaf nodes (i.e., collected samples) and branches of the tree (i.e., correspondence between samples and nodes). Based on this, XGBoost constructs an objective function, which includes a loss function to represent the bias and a regularity term to prevent overfitting. The first-order derivative of the objective function with respect to the number of samples yields the weights of the samples, which are used to evaluate the structure of the tree, with smaller weights indicating a better tree structure. In addition, XGBoost model optimizes the objective function through mathematical methods such as the second order Taylor's formula expansion, and finally obtains a complete prediction model. Because of this, XGBoost is more accurate than gradient boosting decision tree, which only applies a first-order Taylor expansion. Extreme gradient boosting has many advantages in that it can sort the data, reducing the amount of computation, and the computation time is shorter (minutes to hours) even with large datasets (e.g., 1 million samples, hundreds of features). However, the drawback of XGBoost is that such a complex sorting space inevitably takes up a lot of memory.

2.2.7. Neural network

When faced with complex data and highly non-linear tasks, the above models are sometimes insufficient. In such cases, neural networks, with their unique simulation of human brain neurons, become a highly promising modeling option. Neural network effectively overcomes the limitations of traditional models and provide new solution ideas for complex problems [56]. This model simulates the structure and function of neuron network in the human brain, which consists of a large number of interconnected neurons. The essence of neurons is a set of mathematical abstract information. For example, when inputting an array, each number in the array is connected with a corresponding weight value, indicating the connection strength with other neurons. Through the linear combination of input values and weights (weighted summation), and then through the conversion of nonlinear activation function, neurons can achieve the mapping from simple inputs to complex outputs. A typical neural network contains three layers: the input layer, which receives the raw data; the hidden layer(s), which perform feature extraction and transformation; and the output layer, which produces the final result. This layered structure allows neural networks to approximate arbitrarily complex functional relationships by combining simple mathematical operations, such as matrix multiplication and nonlinear function applications. Key nonlinear activation functions introduce nonlinear expressiveness to the model, allowing it to learn and represent complex patterns. Both traditional models and neural networks fit the data by optimizing the objective function. However, neural networks can be considered as an extension of traditional models. While traditional models (e.g., linear regression, decision trees, SVM) usually rely on simple mathematical functions to achieve learned features and pattern recognition, neural networks achieve complex feature learning through multilayer structures and nonlinear transformations.

In the field of biomaterials, the application of neural networks has gradually emerged. Schulz et al. were the first to apply neural network models to material property prediction (Fig. 5). Their research focused on polyethylene/nanoclay composites by collecting data from 100 samples and using neural networks to predict mechanical properties, such as Young's modulus, yield strength, and fracture toughness. The material composition and structural parameters were used as input variables, while the mechanical property indices were the output variables. The trained neural network was able to predict the mechanical properties of the materials accurately, with the prediction error kept within 5% [57]. The power of neural networks lies in their ability to automatically adjust the weighting parameters through the training process, gradually reducing the error between the predicted output and the true value. Also, the model has good fault tolerance and robustness. For instance, in image classification tasks, random noise up to 10–20% usually does not significantly degrade the model accuracy. Neural networks can also process multiple inputs in parallel and compute multiple

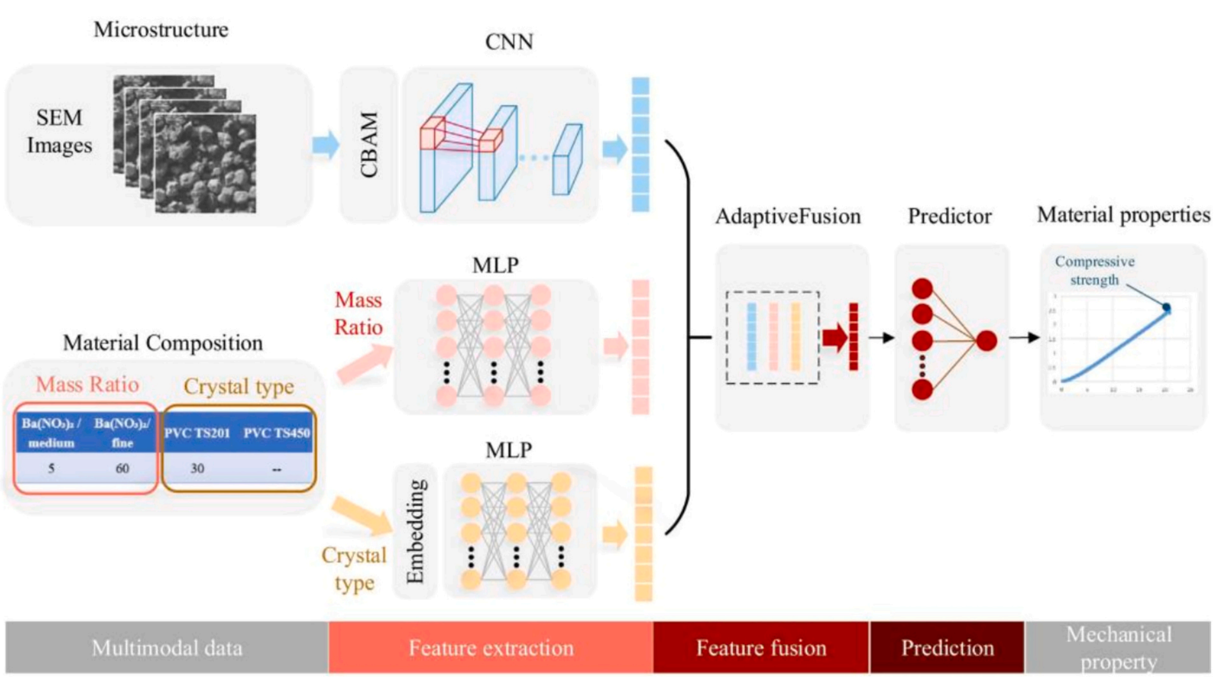

Fig. 5. A framework for performance prediction models of multimodal fusion materials based on neural networks [57].

samples at the same time, which can be accelerated by using the GPU's single-instruction, multiple-data architecture for accelerating matrix operations. For datasets containing 10,000 to 1 million samples, training can typically be completed in hours to days. Only when sample sizes exceed 1 million, training time and hardware consumption increase significantly.

2.2.8. Deep learning

When processing high-dimensional and complex tasks such as image recognition and natural language processing, the feature expression ability of traditional shallow neural networks is gradually showing its limitations due to the small number of layers - which pushes researchers to break through the bottleneck of "depth" by stacking more non-linear transformation layers to build deep architectures, resulting in the production of deep learning. Deep learning is a model based on a simple neural network with a deeper layer. At its core, it uses a neural network with a 'deep' hierarchy to perform tasks that would normally require human intelligence, where the number of hidden layers in the network can be as high as tens of layers or even more. The 'deep' hierarchy refers to the hidden layers of the neural network, which can be tens or even thousands of layers. These layers of neurons are connected to each other, and each neuron layer receives input from the previous layer, mathematically transforms it, and passes it on to the next layer. Layers close to the input layer (usually layers 1–3) are responsible for learning low-level features, while deeper layers (10+) learn more abstract and complex representations (Fig. 6). The relationship between deep learning and neural networks is recursive. Deep learning is essentially an extended form of neural networks. Both are based on mimicking the structure and function of the human brain, processing information through connections between neurons, and adjusting weights to learn patterns and features in data. However, there are significant differences between the two in terms of hierarchical structure and task complexity. Traditional neural networks typically have only a few layers and are capable of handling relatively simple tasks [58]. Deep learning, on the other hand, uses neural networks with multiple layers to perform complex tasks. In deep learning, the more layers a neural network has, the more abstract and complex the features that can be learned.

Applications in the field of biomedical materials have emerged in recent years. In 2021, Lan et al. established an automated deep learning algorithm, the osteogenic convolutional neural network (OCNN), for the quantitative measurement of osteogenic differentiation of rat bone marrow mesenchymal stem cells (rBMSCs) [59]. OCNN was trained using laser confocal scanning microscopy to capture images of rBMSCs with stained cytoskeleton and nuclei at early differentiation stages. OCNN was able to successfully distinguish differentiated cells at early-stage (24 h) with a high area under the curve (AUC) (0.94 ± 0.04) and correlate with traditional biochemical markers. OCNN outperformed individual morphological parameters and SVM in predicting osteogenic differentiation. Moreover, OCNN showed a very strong correlation with the results of Alizarin Red staining after 14 days in predicting the osteogenic induction capacity of small molecule drugs such as simvastatin (Pearson correlation coefficients of 0.9407, 0.9257, and 0.8411, respectively, $p < 0.05$). Similarly, OCNN showed significant correlation in predicting the osteogenic induction capacity of BMP-2 (Pearson correlation coefficient of 0.9090, $p < 0.05$). These results indicate the reliability and feasibility of OCNN in drug screening. Early applications have laid the foundation for subsequent research and dissemination. The design of the multi-layer structure enables deep learning models to extract complex features directly from raw data, without the need for tedious manual feature engineering of traditional methods. Deep learning is essentially an extension and deepening of neural networks, which improves the expressive and learning capabilities of the model by increasing the number of network layers. This characteristic allows it to show significant advantages when dealing with complex data and tasks, especially in feature abstraction and pattern recognition.

3. Applications of machine learning in biomedical nanomaterials

In recent years, machine learning techniques have been reshaping the research paradigm of biomedical nanomaterials design, gradually transforming from experience-driven to data-driven. Their applications can be divided into three main directions: prediction of physical and chemical properties, prediction of synthetic routes, and predictions of interactions between nanomaterials and biological components (Table 3).

3.1. Predicting physical and chemical properties of nanomaterials

The physical and chemical properties of nanomaterials include particle size, surface charge, crystal structure, specific surface area. These properties directly determine the performance of nanomaterials in practical application, including fields such as catalysis and drug delivery. Therefore, studying the physicochemical properties of nanomaterials is rather important. Compared with traditional experimental methods, machine learning models can efficiently process complex data sets, rapidly screen potential material combinations, and reveal hidden constitutive relationships through nonlinear modeling, significantly shortening research and development cycles and reducing experimental costs. The classification of nanomaterials into inorganic and organic categories is justified by fundamental compositional and structural differences that dictate distinct machine learning approaches. Inorganic nanomaterials (non-carbon-based metals, metal oxides,

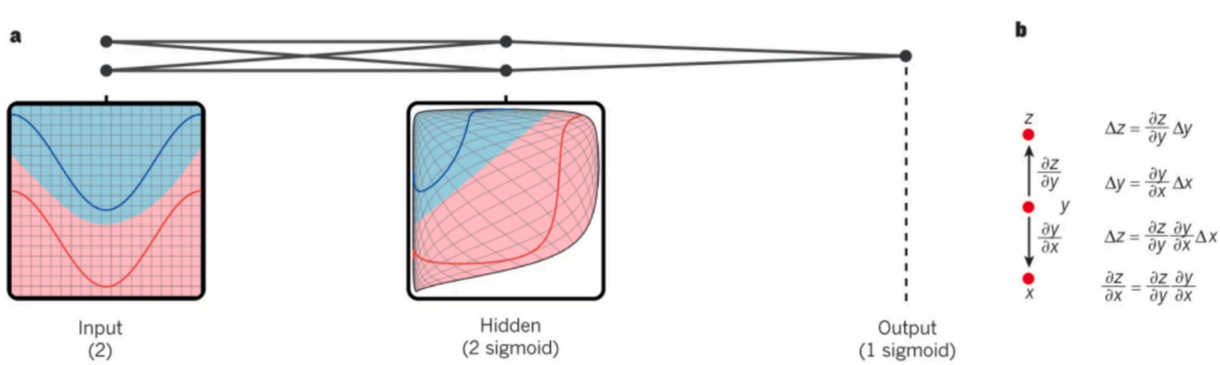

Fig. 6. Multi-layer neural networks and backpropagation [58].
Multi-layer neural networks (represented by connected points) can distort the input space, making data categories (such as the examples on the red and blue lines) linearly separable. The regular grid in the input space is also transformed by the hidden cells (as shown in the middle panel). This is an example that only contains two input units, two hidden units and one output unit, but the network used for object recognition or natural language processing contains tens of thousands of units.

Table 3
An overview of the utilities of different ML models.

	Nanomaterial type	ML model used	Prediction goals and results	Dataset size (Training)	Research references
Predicting physical and chemical properties of nanomaterials	Core-shell nanoparticles	Multi-layer neural network	The resonance of nanoparticles peaks appeared within a specific wavelength range	Over 18,000 samples	[13]
	(1) Diamond nanoparticles (2) Silver nanoparticles	Random forest	Successfully predicted their properties	(1) 500 (2) 425	[14]
	Nano-enzymes	Deep DNN	Reveal the core role of transition metals in determining the catalytic activity of nano-enzymes	920	[15]
	Block copolymer nanoparticles based on polyethylene glycol and polylactic acid	Artificial neural network	The polymer molecular weight had the most significant effect on the particle size, while the polymer-to-drug ratio played a dominant role in the encapsulation rate.	Not mentioned	[16]
	Liposomes	A continuous manufacturing system	Make it more relevant to real-world industrial production scenarios and improve the model prediction accuracy	Not mentioned	[17]
	Single-chain nanoparticles (SCNPs)	A combination of molecular simulation and unsupervised machine learning	Precursor chains with low functionalization ratios exhibited greater size dispersion but relatively homogeneous local environments, whereas precursor chains with high functionalization ratios tended to generate more compact spherical SCNPS	Not mentioned	[18]
Prediction of synthetic routes for nanomaterials	Curcumin-PLGA nanoparticles	A five-layer neural network simulator	Determine the optimal operating parameters, average particle size, yield, and the drug loading capacity	10	[19]
	Polycaprolacto-ne nanoparticles	Neuro Fuzzy Logic (NFL)	Confirm the optimal synthesis conditions	299	[20]
	mRNA vaccine lipid nanoparticles	A machine learning model based on the LightGBM algorithm	Identify the combination that optimized the model performance	325	[21]
	mRNA vaccine lipid nanoparticles	XGBoost and Bayesian optimization techniques	Lipid nanoparticles with particle sizes of about 80 nm and 200 nm were successfully prepared under optimal fabrication conditions	16	[22]
Predicting interactions between nanomaterials and biological components	TiO_2 and ZnO nanoparticles	A model based on multiple linear regression and linear discriminant analysis	Relatively find the key features that decided the extent of cell membrane damage for TiO_2 and ZnO	Not mentioned	[23]
	TiO_2 nanoparticles	Compare linear (multiple linear regression) with nonlinear model (SVM, radial basis function neural network, and generalized regression neural network)	The nonlinear approach has an advantage in dealing with complex mechanisms	31	[24]
	41 types of nanoparticles, including SiO_2, Ni, and NiO	A quantitative structure-toxicity relationship perturbation model based on linear discriminant analysis	Redicted the cytotoxicity of SiO_2 (15 nm and 46 nm), NiO (44 nm) and Ni (65 nm) nanoparticles	Data from 16 studies	[25]
	Gold nanoparticles	A random forest and k-nearest-neighbor algorithm-based quantitative nanostructure-activity relationship model	The model's prediction accuracy exceeded 80%	191	[26]
	PEGylated gold nanoparticles	Deep neural network model	Predicted the blood clearance and organ accumulation of unknown nanoparticles with 77.32%–93.98% accuracy	Not mentioned	[27]
	Inorganic nanoparticles for tumor treatment	Random forest and so on	The size, shape and surface charge of the nanoparticles were found to be the key factors affecting their in vivo therapeutic effect	Data from data from 745 articles	[28]
	A wide range of organic nanoparticles	Decision trees	The model performed well in predicting cytotoxicity, achieving an accuracy of 87.9% ± 2.2%	2896	[29]
	Protein crowns on the surface of nanoparticles	Random forest	Unmodified nanoparticles and surface modifications are key determinants of protein crown formation	652	[30]
	Plasmid DNA lipid nanoparticles	lightGBM model	A molar percentage of charged lipids between 9% and 50%, a molar percentage of PEGylated lipids close to 0.2% to 1%, and an N/P ratio close to 12 significantly improved the transfection efficiency of lipid nanoparticles	1080	[31]
	Nanoparticles in drug delivery to the tumor	Deep neural network model	Positively charged nanoparticles may bind more readily to negatively charged cell membranes, thereby increasing their uptake efficiency in tumors	376	[21]
	Ferritin nanocages	Deep learning model based on image segmentation	He delivery efficiency of the modified ferritin nanocage is significantly improved in low-permeability tumors	More than 67,000	[32]

semiconductors) possess periodic structures requiring descriptors for lattice parameters and electronic properties [60]. Organic nanomaterials (carbon-based liposomes, polymer nanoparticles) are soft-matter systems lacking long-range order, characterized by molecular weight and polymer dynamics. This distinction critically impacts ML modeling: inorganic systems leverage periodic graph representations, while organic systems rely on molecular descriptors [61]. Separating these categories enables feature engineering and model architecture selection, aligning algorithms with underlying material physics for more accurate property predictions.

3.1.1. Prediction of physical and chemical properties of inorganic nanomaterials

The prediction of physicochemical properties of inorganic nanoparticles by machine learning has generally experienced a shift from physical properties to chemical properties to interactions with biologically active molecules, with smaller and finer scales of study.

The earliest research focused on physical properties. In 2019, So et al. proposed a deep learning-based reverse design method to study the optical properties of core-shell nanoparticles, which was used to simultaneously optimize the material and structural parameters of core-shell nanoparticles [24]. This study learned the correlation between the matting spectra of core-shell nanoparticles and design parameters (such as material information and shell thickness, etc.) through a multi-layer neural network, successfully adjusting the material composition of core−shell nanoparticles, so that their resonance peaks appeared within a specific wavelength range. The training dataset contains over 18,000 samples and is proportionally divided into the training set (approximately 80%), the validation set (approximately 10%), and the test set (approximately 10%). To verify the credibility of this prediction, the author used a test set containing 2313 test datasets for calculation and determined that the mean square error (MSE) between the predicted response and the target input spectrum was approximately 9.43×10^{-3}. This quantitative result indicates that the model can indeed provide design parameters that conform to the required optical characteristics. This reverse design method based on deep learning significantly improves the design efficiency. Compared with the traditional genetic algorithm optimization method, it greatly reduces the computing time and cost. Traditional optimization methods require approximately 187 min to find an optimal design, while once this model is trained, it can provide structural design in <1 s.

In 2021, Li et al. proposed a reverse design method based on multi-objective random forests to predict other physical properties, such as formation energy, ionization potential, and electron affinity energy [62]. This study utilized two public datasets: one containing data on 500 diamond nanoparticles and the other on 425 silver nanoparticles. Through features and labels generated by electronic structure simulation, forward and reverse prediction models were constructed. A key advancement in this study over So et al.'s research was the use of Gini importance feature selection. This approach optimized the dataset based on feature importance and reduced the influence of noise, especially in smaller training sets (less than 1000 sample sizes). In the study, the dataset was first optimized through feature selection to remove noise, and the most important features were selected based on the Gini importance of the random forest. This led to 48 features for both the diamond nanoparticle silver nanoparticle datasets. The model successfully predicted the physicochemical properties such as the formation energy, ionization potential, and electron affinity of the nanoparticles. For example, when predicting the properties of the nano-diamond dataset, the model only used nine important structural features. The mean absolute error (MAE) was 0.047 and the MSE was 0.010, indicating that the model can predict physicochemical properties from structural features with high precision.

In addition to physical properties, machine learning can also predict the chemical properties of inorganic nanoparticles. In 2022, Wei et al. studied the enzymatic catalytic activity of nano-enzymes by constructing a deep neural network (DNN) model. Based on the previous DNN model [63], they introduced SHapley Additive exPlanations (SHAP) interpretability analysis to reveal the contributions of various features (such as the influence of metal valence states on catalytic activity). The research team extracted 920 data points from over 300 literatures, covering both internal factors (e.g., metal type, metal valence state, size, shape) and external factors (e.g., buffer pH, temperature, substrate). The dataset was divided into four parts, each used to predict the activity levels of peroxidase (POD), oxidase (OXD), superoxide dismutase, and catalase. By analyzing SHAP values, the study revealed the core role of transition metals in determining the catalytic activity of nano-enzymes. For example, the model predicted that ruthenium-based nanomaterials have a relatively high POD-like activity, while manganese-based nanomaterials exhibit a relatively high OXD-like activity. The experimental verification results are highly consistent with the model prediction.

3.1.2. Prediction of physical and chemical properties of organic nanomaterials

The research on predicting the properties of organic nanoparticles by machine learning models has generally progressed through three stages: using artificial neural networks to predict physical and chemical properties, developing models beyond neural networks, and diversifying the sources of datasets. In 2014, Shalaby's team utilized an artificial neural network model to calculate the particle size and drug encapsulation rate of block copolymer nanoparticles based on polyethylene glycol and polylactic acid [64]. The experimental data covered nanoparticles characterization under 27 different conditions, and two neural network architectures with neuron ratios of 3:10:1 and 3:7:1 for the input, hidden, and output layers respectively. Analysis of the variables revealed that the polymer molecular weight had the most significant effect on the particle size, while the polymer-to-drug ratio played a dominant role in the encapsulation rate. In subsequent years, several other teams also used artificial neural networks to predict the particle size, polydispersity index (PDI), and zeta potential of drug-loaded agar nanospheres [65], chitosan-sodium tripolyphosphate nanoparticles [66], progesterone-loaded solid lipid nanoparticles [67], and PLGA nanoparticles [68,69]. Early research mostly relied on artificial neural networks, but in recent years, other algorithms have been gradually introduced, and the model architecture has evolved from shallow perceptron to deep generative models.

In addition to neural networks, researchers have also tried to develop other machine learning models to predict the properties of organic nanoparticles, hoping to find more models with higher accuracy in specific scenarios. Algorithms such as random forest, XGBoost, light gradient boosting machine (LightGBM), SVM, and multivariate linear regression have been tested, with results proving their accuracy. In 2018, Öztürk et al. used random forests to predict the particle size of solid lipid nanoparticles (SLNs), achieving an MAE as low as 0.028. The experimental data were based on the particle size measurements of SLNs synthesized at different mixing times and formulation components (such as Dynasan 116 and Tween 80 concentrations), and included a total of 45 data points. The research by Öztürk et al. clearly revealed how mixing time and formulation components influenced the particle size of SLNs. Following this study, researchers also applied other algorithms to predict the particle size, zeta potential, encapsulation efficiency, and other key parameters of metal-organic nanocapsules [70], liposomes [71], and PLGA nanoparticles [9,72,73].

In subsequent studies, researchers have found that in order to fit the needs of production practices and reduce cost, datasets from sources other than experiments need to be developed. For example, in 2021, the Sansare team utilized a dataset from a continuous manufacturing system developed at the University of Connecticut, which continuously processes and transforms materials throughout production [74]. While they still predicted the particle size and PDI of liposomes through artificial neural network models, they notably used real-time data from the continuous manufacturing system to construct the training set for the first time, making it more relevant to real-world industrial production scenarios. The study introduced molecular descriptors (e.g. lipid molecular structure parameters) as input features, which significantly improved the model prediction accuracy.

In contrast to the above studies, the studies by Patel et al. and Ayush et al. turned to the use of molecular dynamics simulations to generate virtual nanoparticles libraries, effectively reducing the cost [75,76]. Patel et al. investigated how the sequence pattern of cross-linking groups in the precursor chains of single-chain nanoparticles (SCNPs) affects their shape and dispersion using a combination of molecular simulation and unsupervised machine learning (UMAP algorithm) [75]. The results indicated that precursor chains with low functionalization ratios exhibited greater size dispersion but relatively homogeneous local environments, whereas precursor chains with high functionalization ratios tended to generate more compact spherical SCNPs. In the same year, Ayush et al. combined the convolutional neural network and random forest model to successfully predict the spatial distribution of nanoparticles in polymer nanocomposites by extracting the spatial distribution features through the convolutional neural network and completing the regression prediction with the random forest model [76]. The research focused on compositional parameters such as microscopic phase separation and nanoparticles bridging. The team performed 80 molecular dynamics simulations under isothermal and isobaric conditions for different polymer properties, and each simulation underwent 10^7 steps of equilibrium and production runs to ensure that the system reached thermodynamic equilibrium. The model predicted the nanoparticles radial distribution function with a root mean square error of 0.04 on the training set and 0.06 on the test set, indicating that the model performs well on both datasets.

3.2. Prediction of synthetic routes for nanomaterials

Unlike the prediction of physicochemical properties of nanomaterials, synthetic route optimization requires considering the correlation between reaction conditions and product properties simultaneously. Traditional experimental methods rely on trial-and-error high-throughput experiments, which are time-consuming and inefficient. Machine learning, trained on experimental data, can quickly identify key parameters, narrow down the optimization range, reduce the number of experiments by and significantly improve the accuracy and confidence of synthetic conditions. In this part, the efficiency advantage of machine learning in synthetic route optimization and its contribution to nanomaterials preparation will be explored with specific case studies. The predictive capability of machine learning in synthetic route optimization stems from its capacity to map complex, non-linear relationships between synthesis parameters and product properties within high-dimensional chemical space. Unlike conventional theoretical models that require explicit kinetic or thermodynamic equations, ML algorithms learn implicit correlations directly from experimental data by constructing quantitative representations (descriptors) of reactants, solvents, and reaction conditions [77]. The underlying mechanism involves dimensionality reduction and feature interaction capture—these models decompose multifactorial synthesis problems into weighted combinations of non-linear basis functions, enabling quantification of both individual parameter contributions and higher-order synergistic effects among variables like temperature, precursor concentration, and stabilizer chemistry [78]. Critically, modern explainable AI techniques (e.g., SHAP analysis) now permit mechanistic interpretation of these "black-box" models by attributing predictions to specific input features, thereby revealing the physicochemical driving forces that govern nucleation, growth, and morphology control [79]. This data-driven paradigm effectively decouples predictive accuracy from complete mechanistic understanding, allowing models to leverage latent patterns embedded in historical synthesis outcomes that conventional human-driven analysis cannot readily discern [80].

Machine learning models for predicting the synthetic route of nanoparticles have evolved from simple to complex approaches This progression was mainly reflected in the reaction conditions, from the basic experimental conditions (e.g., the ratio of materials, stirring), to more refined experimental conditions (e.g., the inner diameter of the injection needle), and ultimately focusing on identifying the most critical parameters to achieve a targeted optimization. Earlier studies mostly focused on the optimization of basic process parameters. In 2018, Zabihi et al. pioneered this approach by training a five-layer neural network on injection flow rate, ultrasonic power, and washing parameters for curcumin-PLGA nanoparticle synthesis, achieving sub-50 nm particles with 97.87% yield and over 44% drug loading—demonstrating ML's ability to efficiently navigate high-dimensional condition spaces [81].

The research was extended from optimizing single reaction conditions to synergistically regulating multiple parameters. By combining high-throughput experimental data with machine-learning models, the optimization of the whole process—from static parameters to dynamic conditions—was achieved. Jara's team applied Neuro Fuzzy Logic models to polycaprolactone nanoparticle preparation, analyzing 299 experimental formulations that integrated static parameters (solvent ratios) with dynamic conditions (needle diameter, flow rate) [82] (Fig. 7). Their systematic analysis of variance confirmed the model's ability to predict particle size and zeta potential while identifying optimal stabilizer-geometry combinations that prevented aggregation.

Following this, other researchers have also applied artificial neural networks to optimize the biosynthetic process of nanoparticles. This methodology advanced further with Wang et al., who employed LightGBM algorithms on 325 mRNA vaccine lipid nanoparticle samples, extracting heterogeneous features spanning antigen type, mRNA modifications, and lipid structures [83]. The model achieved correlation coefficients >0.87 on both training and validation sets, revealing that N/P ratios and ionizable lipid selection critically dictate protein expression efficiency.

In recent years, feature engineering has expanded from single experimental parameters to multiple sources of heterogeneous data. The prediction of results has evolved from single-indicator optimization to multi-objective co-optimization. As research becomes more extensive, there are more parameters associated with nanoparticle synthesis routes, necessitating strategies to reduce redundancy in the datasets. Sato et al. integrated XGBoost with Bayesian optimization for mRNA delivery lipid nanoparticles, using only 16 experimental data points to efficiently screen upstream parameters (ethanolamine concentration, buffer pH, flow rates) [84]. This approach successfully targeted specific particle sizes (80–200 nm) with >80% encapsulation efficiency, illustrating how ML-driven adaptive sampling can maximize information gain from minimal experiments.

3.3. Predicting interactions between nanomaterials and biological components

The application of machine learning models in predicting nanomaterial synthesis routes provides important support for the precise preparation of nanoparticles by optimizing synthesis conditions and material selection. The final application of nanomaterials often extends beyond the laboratory environment, especially in the biomedical field, where their practical applications need to consider the complex in vivo environment. Whether predicting the physicochemical properties of nanoparticles through existing literature or optimizing synthesis conditions using experimental data, the ultimate goal is to address one core issue: how to ensure the safety and functionality of nanoparticles in living organisms. Therefore, the interaction between nanoparticles and biomolecules (e.g., toxicity, therapeutic effect, etc.) becomes a key factor determining the success or failure of their application. This area is not only a hot spot for nanotechnology research, but also one of the important research directions for machine learning models.

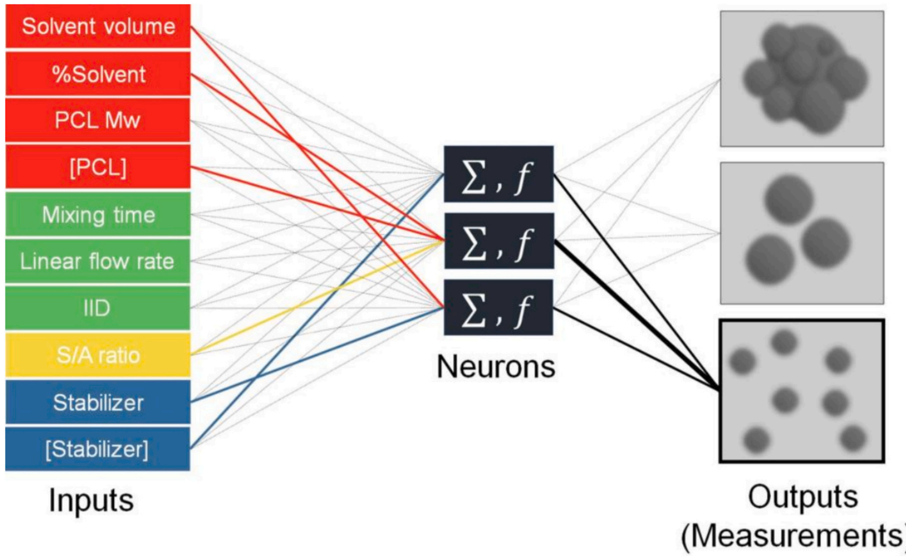

Fig. 7. Schematic diagram of the neural fuzzy logic model [82]. It shows the inputs selected for modeling the measurement output, and these inputs achieve the learning process through neuron "weighted" variables. The colors represent the inputs related to the solvent phase (red), the countersolvent phase (blue), fluid dynamics (green), and the solvent/countersolvent ratio (yellow).

3.3.1. *Toxicity prediction*

Predicting the toxicity of nanomaterials is a critical step in ensuring their safe development and biomedical application, as early identification of potential biological hazards can guide the design of safer nanomedicines and reduce costly late-stage failures. Early studies focused on applying simple models, such as linear models, in conjunction with physical and chemical properties to predict cytotoxicity. For example, Sayes and Ivanov developed linear discriminant analysis models to predict TiO_2 and ZnO nanoparticle-induced cell membrane damage from small datasets (24 and 18 measurements) [85]. Key predictors included particle size, concentration, and zeta potential in water for TiO_2, and size/concentration in culture medium for ZnO. Classification errors of 0 (TiO_2) and 0.111 (ZnO) demonstrated high accuracy. However, the limited sample sizes and purely correlative nature without mechanistic validation constrain the model's broader applicability and generalizability in nanotoxicology. Labouta et al. analyzed 2896 organic nanoparticle samples via decision trees (J48) with feature selection, using training data from 93 publications [86]. Features included material chemistry, size, charge, concentration, cell type, and detection methods. The model achieved 87.9% ± 2.2% accuracy via 10-fold cross-validation, a 15% improvement over the 76.3% baseline. Material chemistry was the primary cytotoxicity predictor, followed by concentration and size.

Due to the complexity of biological interactions, linear models are not always sufficient to predict the interactions between nanoparticles and biomolecules, and the prediction may require nonlinear models. Therefore, research gradually shifted from simple linear regression to the exploration of nonlinear models, such as SVM, neural networks, and so on. Papa et al. compared linear (MLR) and nonlinear models (SVM, RBFNN, GRNN) for TiO_2 nanoparticle cytotoxicity prediction using only 31 data points [87]. Nonlinear approaches achieved superior RMSE values of ~0.1 (SVM, GRNN) and ~0.08 (RBFNN), yet the minimal dataset severely compromises statistical robustness and generalizability. The study also reported a classification tree based on particle size and concentration that predicted cell membrane damage with >80% accuracy, though its reliability remains questionable given these constraints.

Based on the previous work, other researchers sought to integrate more complex machine learning models. However, they found that more complex large-scale machine learning models require large-scale datasets for accurate and reliable predictions. This led to the development of datasets based on literature search and virtual libraries. The advantage of this approach lies in the fact that the data is essentially obtained from actual experiments, which can guarantee the quality and amount of data and save time. Luan et al. developed a QSTR perturbation model via linear discriminant analysis on 16 studies of 41 nanoparticles [88] (Fig. 8). The model calculated 'perturbation terms' (differences in chemical composition, shape, size, exposure time, cell type, cytotoxicity) to reflect toxicity changes under varying conditions. External validation achieved >78.05% accuracy for SiO_2-15 nm, successfully predicting cytotoxicity of silica (15 and 46 nm), nickel oxide (44 nm), and nickel (65 nm). Huang et al. developed machine learning models predicting metal oxide nanoparticle (MeONP) cytotoxicity in THP-1 macrophage cells, achieving 97% training and 96% test accuracy with a consensus model (RF + k-NN + DT) after SMOTE balancing [89]. The study identified lysosomal dissociation degree (PSF, pH 4.5) as the primary toxicity determinant, validated via confocal microscopy showing MeONP-lysosome colocalization. Key descriptors included ζ-potential, metal electronegativity, primary size, and concentration. External validation with four novel MeONPs demonstrated 91% accuracy, establishing a robust framework for nano-safety assessment and safer design.

Other researchers have also applied Bayesian linear models, quantitative structure-toxicity relationship perturbation models, Bayesian regularized artificial neural networks, genetic algorithms, and categorical boosting algorithms to predict the cytotoxicity of various nanoparticles, including iron oxides [90], gold nanoparticles [91], ZnO nanoparticles [92], silver nanoparticles [93], bimetallic nanoparticles [94].

While significant progress has been made in predicting cytotoxicity using increasingly sophisticated models, a critical frontier remains largely uncharted: expanding the predictive framework to encompass deeper and more complex toxicities. Future research must extend beyond cytotoxicity to address the prediction of immunotoxicity, genotoxicity, and reproductive toxicity, which are essential for a comprehensive nanosafety assessment but currently lack robust, data-driven predictive models. Developing such capabilities will require the generation of high-quality, standardized datasets specifically tailored for these endpoints and the application of advanced machine learning techniques capable of unraveling their underlying biological mechanisms.

3.3.2. *Protein binding prediction*

It is critical to evaluate the fate of nanoparticles in vivo such as

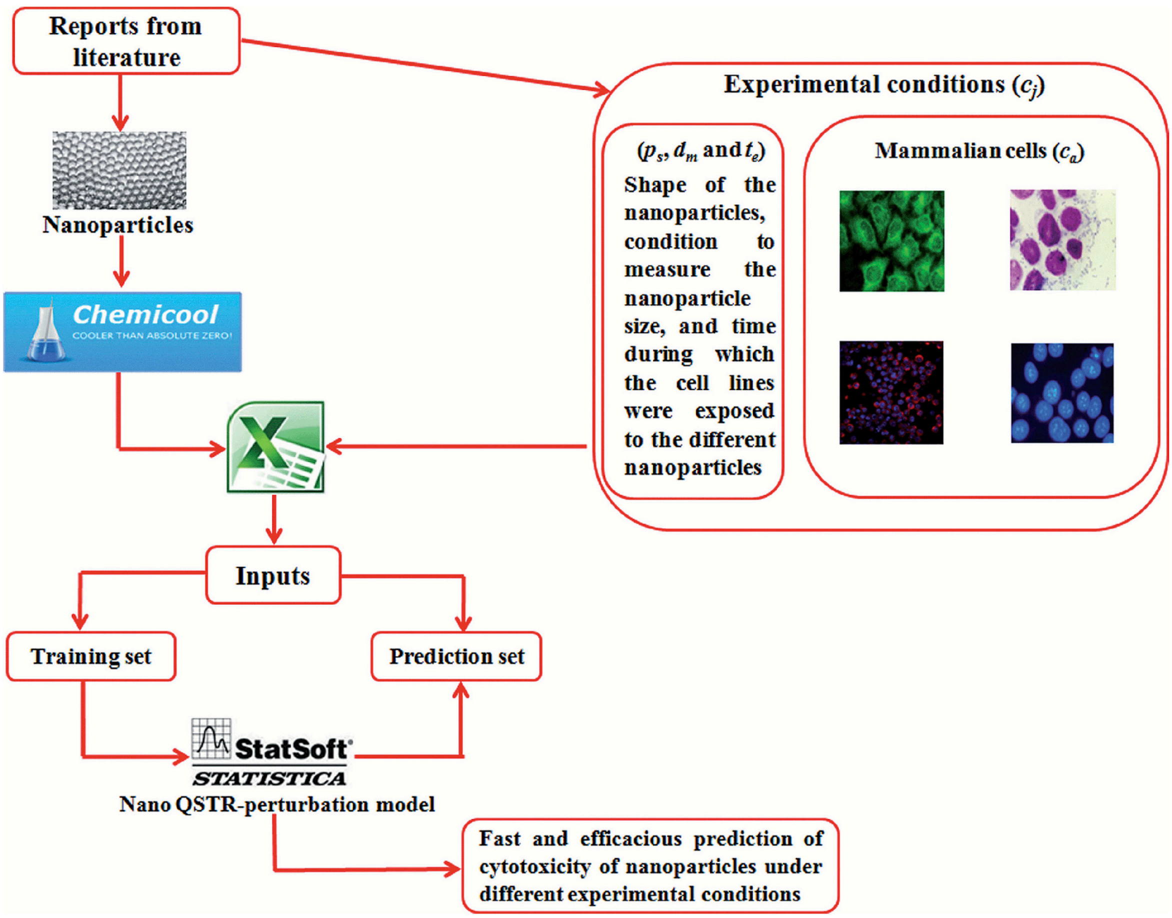

Fig. 8. The general steps adopted for establishing the perturbation model of the quantitative structure-toxicity relationship [88].

nanoparticle retention and metabolism, to assess their biosafety. Predicting the in vivo fate of nanoparticles is more complex than predicting cytotoxicity because it involves multiple organs and protein interactions and requires better model performance. Lazarovits et al. developed a deep neural network model to predict the in vivo fate of PEGylated gold nanoparticles, including blood clearance and liver/spleen accumulation [95] (Fig. 9). After injecting nanoparticles in rats, they performed mass spectrometry at multiple time points to quantify surface-adsorbed proteins. Input features included label-free protein intensity, while outputs comprised clearance, organ accumulation, and particle size. The model achieved 77–94% accuracy via K-fold cross-validation with early stopping. The study revealed that nanoparticle scavenging depends on combined patterns of multiple surface proteins rather than individual proteins.

Virtual libraries using procedurally generated nanoparticles can also serve as databases for machine learning. Unlike various forms of experimental data, virtual libraries contain significantly more features than experiments, with each type of nanoparticle having a corresponding value with no missing values. The sample size can be large or small, depending on the need. Yan et al. proposed a random forest and k-nearest-neighbor algorithm-based quantitative nanostructure-activity relationship model to predict the biological activity of gold nanoparticles [96] (Fig. 10). Trained on 191 GNPs with 126 geometric descriptors, it achieved >80% accuracy for cellular uptake, ROS, and AChE binding. The CNOM descriptor predicted elevated ROS for amide/amine-functionalized particles. While the virtual library construction was innovative, the description lacks critical analysis of descriptor biological meaning, validation robustness, and model limitations, offering limited translational insight.

Protein crowns, a layer of proteins adsorbed on the surface of a nanoparticle after it enters a biological environment, can alter the surface properties of the nanoparticle, affecting its interactions with the cell and reducing its targeting properties. Researchers employed machine learning models to probe the influences on protein crown formation and even cellular selectivity. A study harnessed random forest with SHAP analysis to model protein corona formation and cell recognition using 652 data points from 56 papers [97]. The model achieved >0.7 correlation coefficients, identifying surface modifications and unmodified nanoparticles as key determinants. While successfully predicting macrophage uptake and cytokine release, this approach predominantly confirms established relationships rather than revealing novel mechanisms. Furthermore, aggregating heterogeneous literature data risks introducing publication bias and methodological inconsistencies, potentially limiting the model's generalizability beyond the training dataset. Cheng et al. robustly optimized cell type-preferential plasmid DNA transfection by applying LightGBM modeling to 1080 lipid nanoparticle (LNP) formulations from high-throughput screening [98]. Integrated SHAP analysis elucidated critical composition-function relationships, revealing that charged lipids at 9–50%, PEGylated lipids at 0.2–1%, and N/P ratio ~12 significantly enhanced transfection efficiency across diverse cell lines. Through systematic feature optimization, average prediction errors were reduced to 5–10%. This work demonstrates how integrating machine learning with high-throughput screening can accelerate rational LNP design by providing quantitative, predictive guidelines that transform traditional empirical formulation development into an efficient, data-driven process.

3.3.3. Targeting ability prediction

The targeting ability of nanomaterials is a crucial prerequisite for achieving diagnostic and therapeutic purposes. For predicting targeting

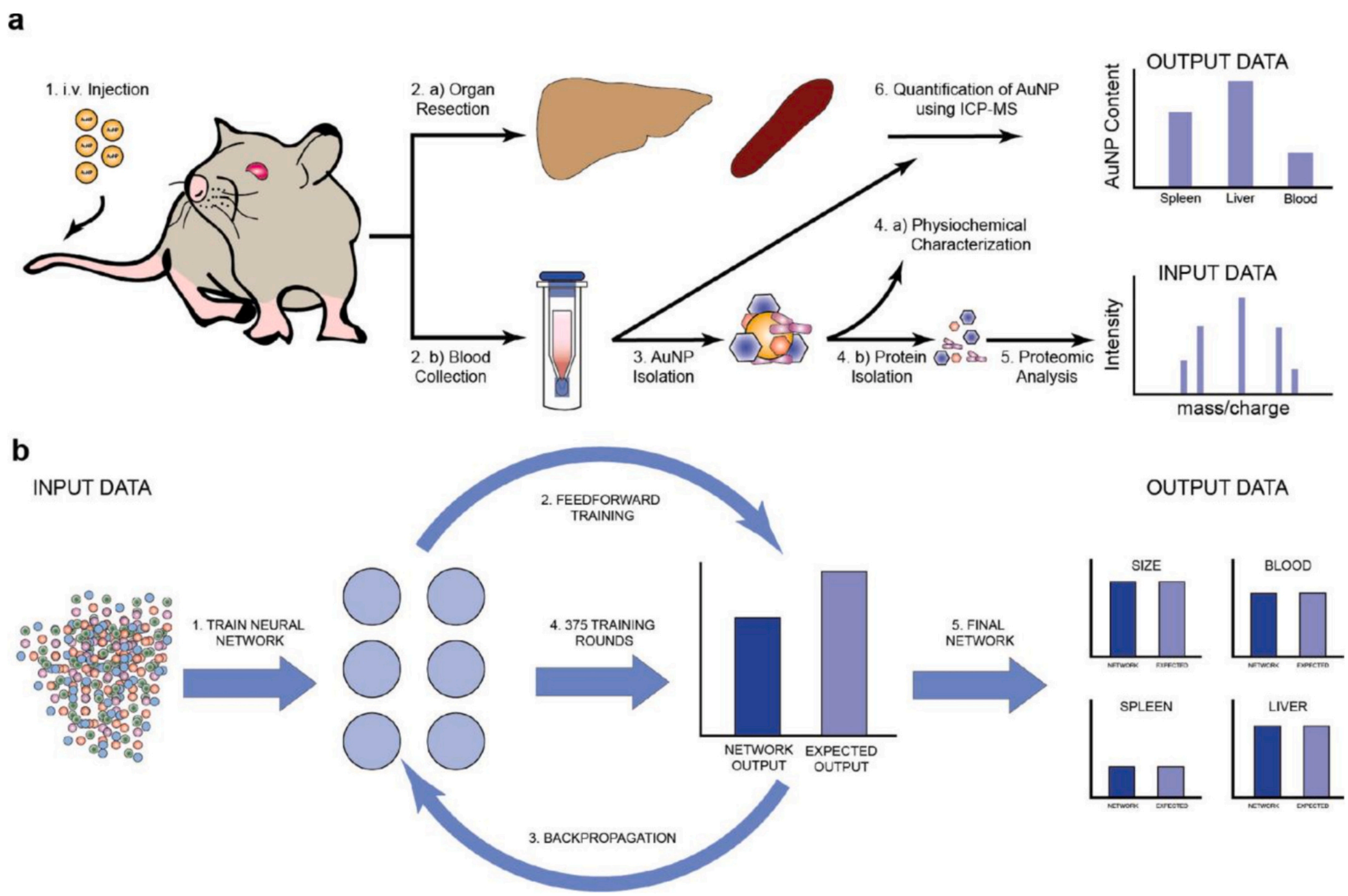

Fig. 9. Methods for constructing and training predictive neural networks [95].
(a) PEGylated AuNPs were injected intravenously into rats, and blood was sampled at multiple times over 24 h from the same animal. AuNPs suspended in serum were separated using centrifugation and underwent physiochemical characterization. Proteins were stripped off and digested for LC–MS/MS processing and analysis. This proteomic analysis formed the input for the neural network (NN). On the other hand, liver and spleen were also resected from rats and digested to measure AuNP organ accumulation using ICP–MS. The organ accumulation of AuNPs and the size of the nanoparticle formed the output of our NN.
(b) NN inputs the protein data and iterates it through a feedforward and backpropagation cycle to build a mathematical model for predicting the output parameters.

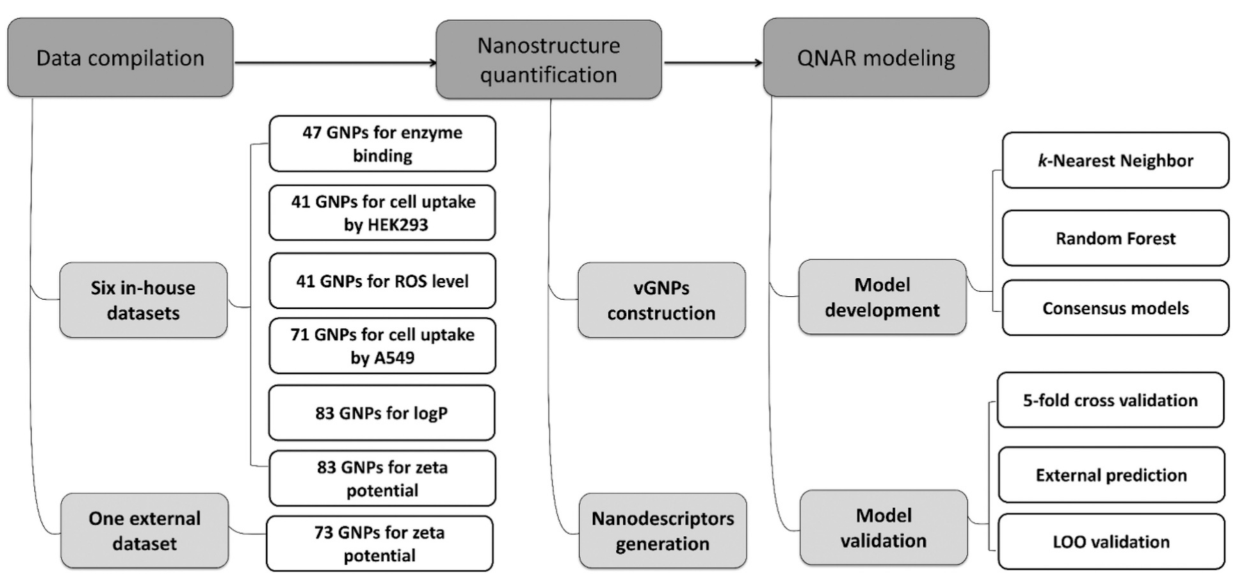

Fig. 10. Schematic diagrams of data compilation, development of virtual gold nanoparticles and modeling of quantitative nanostructure-activity relationship models [96].

ability, image data are becoming a novel type of training set material that overcomes the limitations of traditional descriptors, saving significant effort in dataset preprocessing while also imposing higher demands on the learning capabilities of models. Zhu et al. employed deep learning-based image segmentation to analyze >67,000 tumor vessels, predicting vascular permeability of ferritin nanocages [99]. Their model demonstrated improved delivery efficiency in low-permeability tumors while confirming passive extravasation dominance in high-permeability ones. Transmission electron microscopy validated these differences across tumor models, informing nanodrug delivery design. Notably, this work pioneered direct image data input for multimodal learning and offers open-source resources. However, the study's generalizability across diverse cancer types and clinical translation pathways remain underexplored, limiting its immediate impact. In addition, researchers used models such as Bayesian ridge regression, kernel ridge regression, K nearest neighbor, artificial neural network,

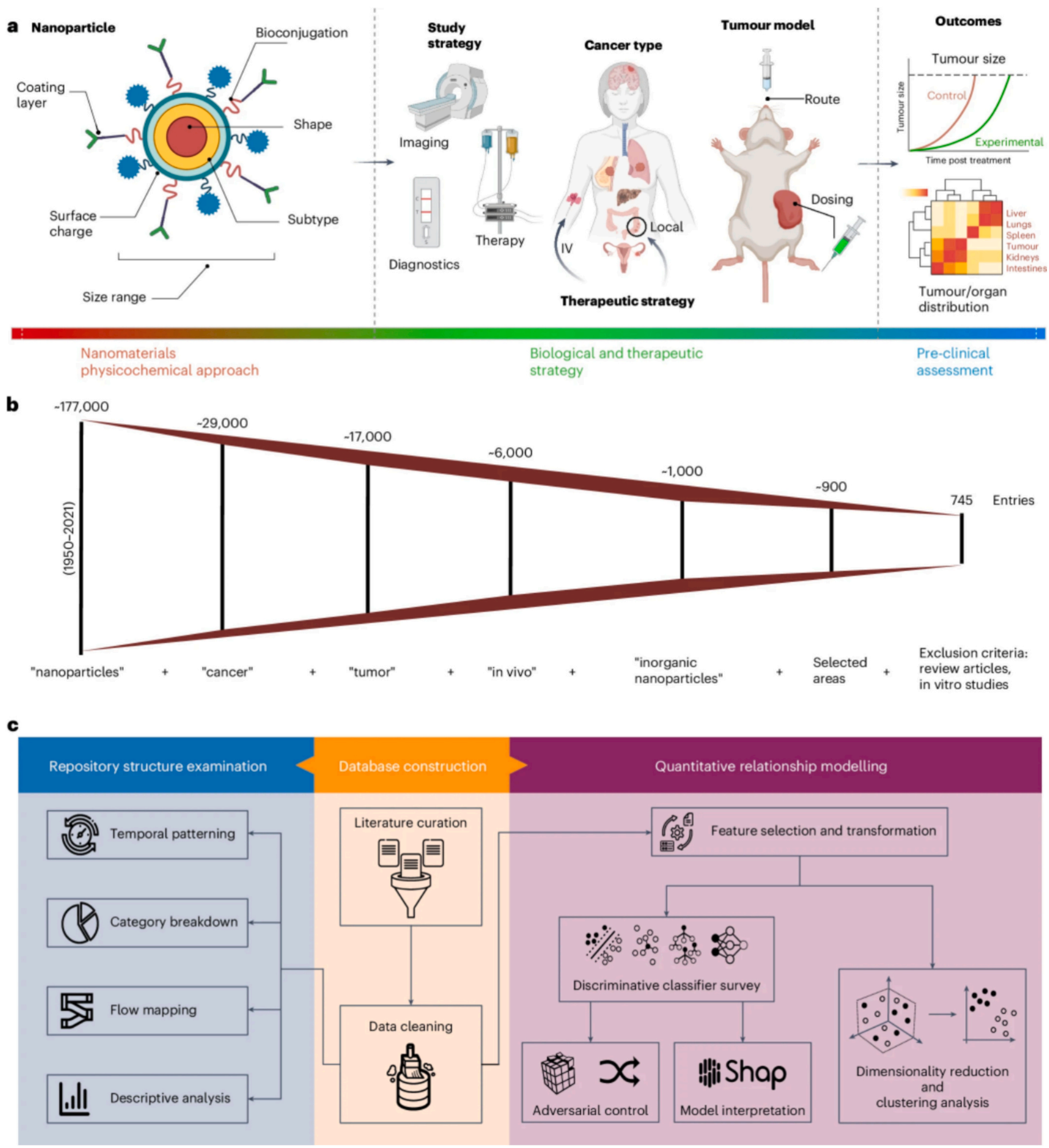

Fig. 11. Artificial intelligence modeling for research design, meta-analysis and workflow [111].
(a) Schematic representation of the experimental design of nanoparticle fabrication and its further biomedical applications; (b) Database curation strategies; (c) Workflow of the analysis, including data curation and preparation, statistical analysis and ML-driven modeling.

and random forest to predict the biodistribution of nanoparticles in different organs (e.g., tumor, heart, liver, spleen, lungs, and kidneys) [100], the release rate of curcumin from lecithin/chitosan nanoparticles [101], the efficiency of trans-placental transfer of lipid nanoparticles [102], and the efficiency of delivered mRNA expression [103].

3.3.4. Drug delivery efficiency prediction

Nanoparticle-mediated drug delivery is another important area of research where machine learning models are incorporated. These models are trained by constructing correlations between the composition and physicochemical properties of nanoparticles and the properties of biomolecules and biological tissues, so that the models can predict what factors influence delivery efficiency. Lin et al. modeled nanoparticle drug delivery using a deep neural network on 376 datasets from the Nano-Tumor database [104]. Their model identified a critical charge-dependent trade-off: moderate positive charge enhances tumor uptake via membrane binding, but excessive charge triggers protein corona formation, reducing efficiency. Despite strong training performance ($R^2 = 0.92$), the substantial test set drop ($R^2 = 0.70$) reveals overfitting risks - an inherent limitation when relying solely on conventional physicochemical descriptors without incorporating structural or dynamic properties, restricting broader applicability across diverse tumor microenvironments. Kumar et al. applied interpretable ML to design gene editing vectors, screening 43 copolymers to identify P38 with 60% efficiency surpassing commercial reagents [105]. Random forest analysis mechanistically linked hydrophobicity and Hill coefficient to RNP release, while demonstrating that multimer size and protonation minimally impact cytotoxicity. However, both studies exclusively employ traditional descriptors, overlooking opportunities to integrate molecular fingerprints or domain-specific features that could improve predictive accuracy and yield deeper mechanistic understanding. This descriptor limitation constrains model generalizability and design space exploration, underscoring the urgent need for feature engineering innovation in future nanomedicine ML applications to overcome these fundamental constraints.

Later, it was shown that non-linear models have limitations, such as poor interpretability, unknown basis of model decisions, and lack of detailed elaboration of internal mechanisms, which affects the verification of accuracy. Mendes et al. constructed a database using 745 articles from the PubMed literature database, and subsequently applied multiple algorithms such as random forest to study the design of inorganic nanoparticles for tumor treatment [106] (Fig. 11). The dataset contains a wide range of inorganic nanoparticles, including gold nanoparticles, iron oxide nanoparticles and mesoporous silicon nanoparticles, characterized by physicochemical properties (e.g., size, shape and surface charge), therapeutic efficacy (including tumor inhibition rates and off-target accumulation). The size, shape and surface charge of the nanoparticles were found to be the key factors affecting their in vivo therapeutic effect, and the modeling results were further validated by SHAP value analysis. The prediction accuracy of the random forest model was verified to be 88% by SHAP value analysis. Specifically, it was found that the shape of the nanoparticles had a significant effect on their therapeutic effectiveness, with rod-shaped nanoparticles being more effective than spherical nanoparticles. In addition, treatments combining photothermal therapy and photoacoustic imaging showed higher therapeutic efficacy. In summary, the study highlighted that size, shape, and surface charge are key factors influencing the therapeutic efficacy of inorganic nanoparticles. Rod-shaped nanoparticles and combined photothermal/photoacoustic treatments were found to be particularly effective.

4. Outlook

This review systematically summarizes machine learning models and analyzes the core principles, advantages, and limitations of unsupervised and supervised learning models. The applications of machine learning models in predicting physicochemical properties, synthesis route, and interactions with biological components were also sorted out and analyzed. The application of machine learning models in predicting inorganic physicochemical properties demonstrates the modeling process that gradually deepens from predicting physical properties to chemical properties. In the field of organic materials, there is a trend from using basic neural network models to applying more complex and diverse models, and experimental data and virtual library generation technology are gradually being integrated to expand the data sources. The existing model can reverse-derive the composition and structural parameters of nanomaterials that meet the requirements based on preset performance, and achieve precise prediction of physical and chemical properties. The research context of machine learning models predicting the optimization of nanomaterial synthesis paths has gradually advanced from a single reaction condition to multi-parameter collaboration. Currently, existing models can handle a large number of mutually coupled reaction conditions involved in the synthesis path. The research on modeling the interaction of biological components in machine learning has undergone a transformation from applying simple linear models combined with limited features to processing large-scale data with complex nonlinear models. The research content has gradually delved from predicting in vitro cellular effects to predicting complex in vivo fates, and some studies have expanded to exploring new descriptors. The existing models are currently capable of cell recognition and the influence of biological effects, and can also predict the dynamic processes of nanomaterials in organisms.

However, despite the remarkable progress made by machine learning in the field of biomedical nanomaterial design, due to the differences in experimental conditions and data collection, many studies have problems such as feature loss and excessive noise in the dataset, which cause great instability to the prediction results. Therefore, there is still a long way to go before using machine learning to precisely guide the synthesis of biomedical nanomaterials and apply them in clinical practice.

4.1. Machine learning methods for transparent decision-making

Although complex models such as deep learning have demonstrated strong capabilities in predicting the properties of nanomaterials, their decision-making processes are often regarded as "black boxes", lacking clear interpretability. This inherent unexplainability seriously hinders researchers from deeply understanding the physicochemical essence and mechanism of action behind the properties of materials. Meanwhile, the effective integration of cross-scale data faces significant challenges, especially in developing the effective correlation between the microscopic scale (such as atomic-level structural information revealed by molecular dynamics simulations) and the macroscopic scale (such as biocompatibility experimental data). Therefore, it is crucial to establish machine learning methods with high interpretability. Design descriptors directly related to the physical and chemical nature of the material (such as band structure parameters, reactive site density, solvation energy) as model input. The model makes predictions based on these features with clear physical significance, and the results are easier to understand and trace. Integrating known physical laws, chemical principles or kinetic equations directly as constraints into the architecture of machine learning models enables the model predictions to conform to physical rules and enhances interpretability. It is also possible to consider using techniques such as SHAP and local interpretable model-agnostic explanations (LIME) to quantify the contribution of each input feature to the final prediction result and reveal the key influencing factors.

4.2. Standardizing nanomaterial data: physicochemical properties and therapeutic effects

One significant challenge currently faced by the researchers lies in the lack of a unified standard for reporting data on the physicochemical properties of nanomaterials and their therapeutic effects on diseases across different articles. This severely restricts the repeatability of the research, the comparative analysis across studies, and poses obstacles to the machine learning models trained using these data. Due to the differences in experimental plans and detection techniques, the quantification methods and presentation forms of the same type of data by different research teams often vary greatly. For example, in the evaluation of the targeting effect, some studies use fluorescence intensity, while others use absolute or relative quantitative units such as %ID/g (percentage of injection dose per gram of tissue). This inconsistency makes it almost impossible to directly compare the data across different literature. Similar problems are also widespread in the reports of treatment efficacy data (such as tumor suppression rate, survival period, etc.). Some research articles fail to fully report the key experimental data. The absence of this information not only leads to incomplete learnable materials for the model but also limits the accuracy of the prediction. Therefore, one of the important directions in the future is to establish and promote a standardized data reporting framework, which should clearly stipulate the material characteristics that need to be included when publishing research articles, as well as the requirements for quantification using recognized numerical units.

4.3. Integrating machine learning with theoretical simulations

Combining machine learning with traditional physical theory simulation methods, such as molecular dynamics simulation, demonstrates great potential and promising prospects. A key convergence point lies in that physical simulations such as molecular dynamics can generate large-scale “virtual material databases” with complete descriptors. These simulations are based on physical and chemical principles and can invoke various data and information at the atomic and molecular levels, providing valuable information sources for material systems that lack sufficient experimental data. These virtual data can greatly enrich the training set of machine learning models, effectively make up for the incompleteness of experimental data, and cover a wider range of material types. As demonstrated by the Open Catalyst Project, training machine learning models using massive molecular configuration data generated by physical simulation can predict catalytic reactions and screen candidate materials more efficiently than traditional methods. Combining the virtual data generated by physical simulation with actual experimental data to construct a more comprehensive and reliable hybrid database is a feasible idea. The ability to generate virtual databases through physical simulation, combined with machine learning, compensates for their respective deficiencies and can achieve greater synergy and efficiency.

4.4. Multi-angle verification of machine learning results to enhance credibility

Although machine learning has been developed and studied to a certain extent in the field of biomedical nanomaterial design, reports on the in vivo verification of new materials in current research articles are still relatively scarce. It is necessary to systematically carry out animal experiments based on the prediction results of the model to verify the accuracy of the predictions. For application scenarios such as drug delivery systems, animal models can directly evaluate core indicators predicted by the model, such as the biocompatibility of nanomaterials, in vivo drug release kinetics, targeting efficiency, and therapeutic effects. These experimental data at the in vivo level are not only the “gold standard” for evaluating the accuracy of the model, but also provide indispensable references for subsequent research. In addition, promote communication and cooperation among researchers in the fields of materials science, computer science and biomedicine, with a focus on exploring the common differences between model predictions and actual animal experiment results and their root causes (such as the influence of complex physiological environments in vivo, the limitations of simplified model assumptions, cross-species differences, data standardization issues, etc.). This cross-border dialogue is crucial for understanding model biases, jointly formulating more effective validation strategies, and optimizing experimental designs to obtain model-friendly data.

Machine learning has a bright future in biomedical nanomaterials. With technological advances and better data quality, it will excel in the following aspects. (1) Optimizing nanomaterial physicochemical property prediction for more accurate disease diagnosis and treatment. (2) Combining with traditional physical simulations to create comprehensive virtual databases. This enriches training data and accelerates the discovery of new materials. (3) Strengthening cross-disciplinary cooperation, especially in validating machine learning predictions. Animal experiments will enhance model credibility and reliability. (4) Advancing data reporting and management standardization. This ensures comparability and integrity across studies, offering more stable and richer data support. Despite challenges, machine learning will keep aiding biomedical nanomaterial design, driving innovation and clinical translation in the field.

4.5. Ethical implications of ML-based nanomaterial safety prediction

The integration of machine learning into nanomaterial safety assessment raises critical ethical considerations that extend beyond technical performance.

A primary ethical advantage of ML-based nanotoxicology is its alignment with the 3Rs principle (i.e. replacement, reduction, and refinement) in animal research [107]. Chen et al. emphasize that deep learning networks can predict nanoparticle toxicity from extensive datasets, providing an efficient and ethically superior method compared to traditional in vivo testing [108]. However, this should not lead to complete abandonment of experimental validation, as in vitro and in silico models may exhibit lower predictive values and cannot fully capture complex organism-level responses [107].

The quality and representativeness of training data pose significant ethical concerns. Existing datasets often suffer from feature loss, excessive noise, and systematic biases due to inconsistent experimental conditions, which may disproportionately affect risk assessments for certain populations or environmental contexts [109]. Without standardized protocols and comprehensive data collection, ML models risk perpetuating health disparities by underpredicting toxicity for nanomaterials prevalent in underserved communities.

The translation of ML predictions into binding safety standards demands robust ethical and regulatory frameworks. Gajewicz-Szter et al. argue that the development of comprehensive strategies and standard controls for ML in environmental risk assessment is crucial to ensure accuracy, reliability, and accountability. Similarly, frameworks promoting the ethical application of AI in related fields like drug discovery highlight the need for governance structures that balance innovation with public health protection [110]. Human expert oversight must remain central to prevent over-reliance on algorithmic decisions that could compromise safety.

5. Methods

5.1. Literature search strategy

A comprehensive systematic search was conducted across Web of Science. The complete search strategy utilized in Web of Science was: TS = (“machine learning” OR “Principal Component Analysis” OR “PCA” OR “K-means” OR “Hierarchical Clustering” OR “Autoencoder” OR “Variational Autoencoder” OR “VAE” OR “Linear Regression” OR “support vector machine” OR “SVM” OR “XGBoost” OR “extreme gradient boost” OR “decision tree*” OR “naive bayes” OR “Random Forest” OR “Neural Network” OR “Deep Learning”) AND PY = (“2000–2025”) AND TS = (“nanomaterial” OR “nanoparticle” OR “nanomaterials” OR “nanoparticles”) AND TS = (“biomedical” OR “biomedicine”). The search was restricted to publication years 2000–2025 with no language filters applied during the initial retrieval.

5.2. Eligibility criteria

Studies were included if they met the following criteria: (1) original research articles employing machine learning methodologies to analyze nanomaterials for biomedical applications; (2) published between January 1, 2000, and December 31, 2025; (3) peer-reviewed journal articles. Exclusion criteria comprised: review articles, editorials, conference abstracts, book chapters, patents, and non-English publications.

5.3. Study selection process

The systematic screening process followed PRISMA 2020 guidelines. The initial search yielded 412 records across all databases. Following automated removal of review articles, conference proceedings, and editorial materials, 297 articles remained for further screening and research.

CRediT authorship contribution statement

Qingya Zhu: Writing – review & editing, Writing – original draft, Conceptualization. **Shufan Feng:** Writing – review & editing. **Yuan Feng:** Writing – review & editing. **Chenhao Ma:** Writing – review & editing. **Zijie Chu:** Writing – review & editing. **Katherine Tang:** Writing – review & editing. **Chun Mao:** Writing – review & editing, Conceptualization. **Ya Guan:** Writing – review & editing, Conceptualization. **Sharon Gerecht:** Writing – review & editing, Conceptualization. **Mimi Wan:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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